

A Product Formula and All-Degree Positivity for the (6, (0, 1))-Core Bipartition Series

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Abstract

Gerber and Norton reduced the remaining level-two case at $e = 6$ to coefficient positivity for two charged-core bipartition series. We prove the entire charge $(0, 1)$ branch. More precisely, we identify its abacus theta series with an eta quotient by a weight-5 Sturm comparison at level 432, and obtain

$$c_{6,(0,1)}(q) = \frac{(q^2; q^2)_\infty^3 (q^6; q^6)_\infty^2 (q^{12}; q^{12})_\infty^2}{(q; q)_\infty^2} = \psi(q)^2 c_3(q^2) \psi(q^6).$$

An application of Legendre's three-square theorem then shows that every coefficient is positive. No positivity result for the charge $(0, 3)$ series is obtained here, so the original conjunction of the two branches remains unresolved.

1 Statement and context

For a partition λ and an integer s , write

$$S(\lambda, s) = \{\lambda_k - k + 1 + s : k \geq 1\} \subset \mathbb{Z}.$$

For the charges used below, a bipartition $\lambda = (\lambda^1, \lambda^2)$ is an $(e, (s_1, s_2))$ -core precisely when

$$S(\lambda^1, s_1) \subseteq S(\lambda^2, s_2) \subseteq S(\lambda^1, s_1 + e). \quad (1.1)$$

We use the generating function

$$c_{e,s}(q) = \sum_{\lambda} q^{|\lambda^1| + |\lambda^2|},$$

where the sum is over the bipartitions satisfying (1.1). These conventions are those of Gerber and Norton [1].

Let

$$(a; q)_\infty = \prod_{m \geq 0} (1 - aq^m), \quad \phi(q) = \sum_{x \in \mathbb{Z}} q^{x^2}, \quad \psi(q) = \sum_{r \geq 0} q^{r(r+1)/2},$$

and let $c_3(q)$ be the ordinary 3-core partition generating function. Gerber and Norton obtain

$$\phi(q) c_3(q)^2 = c_{6,(0,3)}(q) + 2q c_{6,(0,1)}(q) \quad (1.2)$$

in their equation (5.3), and list the separate positivity of the two $e = 6$ series as conjectural [1, Conjecture 3.10 and Table 2]. Our result settles exactly one of those two statements.

Theorem 1.1. *As an identity of formal power series,*

$$c_{6,(0,1)}(q) = \frac{(q^2; q^2)_\infty^3 (q^6; q^6)_\infty^2 (q^{12}; q^{12})_\infty^2}{(q; q)_\infty^2} = \psi(q)^2 c_3(q^2) \psi(q^6). \quad (1.3)$$

Consequently,

$$[q^n] c_{6,(0,1)}(q) > 0 \quad \text{for every } n \in \mathbb{Z}_{\geq 0}.$$

Exact finite certificates can give rigorous theorems inside a prescribed finite model while leaving the ambient problem explicitly open; see Zeng et al. [5] for two examples in finite discrete geometry. Here the finite Sturm comparison instead certifies an identity of modular forms, and the all-degree positivity conclusion comes from the ensuing analytic factorization. The cited finite-geometry results are not used in the proof below.

2 From charged abaci to six lattice cosets

Put $A = S(\lambda^1, 0)$ and $B = S(\lambda^2, 1)$. We index the six abacus runners by $i = 1, \dots, 6$, with the sixth runner representing the zero residue class.

Lemma 2.1 (Exact abacus parametrization). *The $(6, (0, 1))$ -core bipartitions are in bijection with pairs*

$$(a, j), \quad a = (a_1, \dots, a_6) \in \mathbb{Z}^6, \quad \sum_{i=1}^6 a_i = 0, \quad 1 \leq j \leq 6.$$

The size of the bipartition corresponding to (a, j) is

$$E(a, j) = 6 \sum_{i=1}^6 a_i^2 + 2 \sum_{i=1}^6 i a_i + 6 a_j + j - 1. \quad (2.1)$$

Proof. The containments $A \subseteq B \subseteq A + 6$ imply $A \subseteq A + 6$. Thus A is the beta set of a charge-zero 6-core. If its first gap on runner i is

$$g_i = i + 6a_i,$$

then charge zero is equivalent to $\sum_i a_i = 0$, and the usual abacus size formula is

$$|\lambda^1| = 3 \sum_i a_i^2 + \sum_i i a_i. \quad (2.2)$$

Moreover, $(A + 6) \setminus A = \{g_1, \dots, g_6\}$. Since B has charge one, it lies between A and $A + 6$ if and only if $B = A \cup \{g_j\}$ for a unique j . Adjoining a beta number g_j to a charge-zero beta set changes the partition size by $g_j - 1$. Consequently

$$|\lambda^1| + |\lambda^2| = 2|\lambda^1| + g_j - 1,$$

which is (2.1). Conversely, every displayed pair constructs nested beta sets of the required charges, so the parametrization is bijective. $\square$

Let

$$A_5 = \{z \in \mathbb{Z}^6 : \sum_i z_i = 0\}, \quad r = (-3, -2, -1, 0, 1, 2), \quad v_j = r + 3e_j,$$

and define

$$\Theta_{6A_5+v_j}(\tau) = \sum_{z \in 6A_5+v_j} q^{(z,z)/2}, \quad q = e^{2\pi i \tau}.$$

Lemma 2.2. *If $C(q) = c_{6,(0,1)}(q)$, then*

$$T(\tau) := q^5 C(q^3) = \sum_{j=1}^6 \Theta_{6A_5+v_j}(\tau). \quad (2.3)$$

The six cosets in this sum are pairwise distinct.

Proof. For a pair (a, j) from Lemma 2.1, put $z = 6a + v_j$. Then $z \in A_5$, and direct expansion of (2.1) gives

$$3E(a, j) + 5 = \frac{1}{2} \sum_{i=1}^6 z_i^2.$$

This proves (2.3). If $j \neq k$, then $v_j - v_k = 3(e_j - e_k) \notin 6A_5$, proving distinctness. $\square$

3 The modular-form identity

We next identify the theta sum in (2.3). Set

$$G(\tau) = \frac{\eta(6\tau)^3 \eta(18\tau)^2 \eta(36\tau)^2}{\eta(3\tau)^2}. \quad (3.1)$$

Proposition 3.1. *One has $T(\tau) = G(\tau)$.*

Proof. We first place both squares in the same integral-weight modular-form space. In the basis $e_i - e_6$ of A_5 , its Gram matrix is

$$A_0 = I_5 + J_5, \quad \det A_0 = 6, \quad A_0^{-1} = I_5 - \frac{1}{6} J_5.$$

Let v'_j denote the first five coordinates of v_j . In the elementary theta notation used in the transformation theorem of Kane and Kim [2, Proposition 2.2], take

$$A = 36A_0, \quad N = 216, \quad H_j = 36v'_j. \quad (3.2)$$

Then $NA^{-1} = 6I_5 - J_5$ is integral, $AH_j \in N\mathbb{Z}^5$, and the substitution $X = 36(z_1, \dots, z_5)^t$ gives exactly

$$\frac{X^t A X}{2N^2} = \frac{(z, z)}{2}.$$

Thus there is no rescaling of the exponent in (2.3).

For $\gamma = \begin{pmatrix} p & b \\ c & d \end{pmatrix}$ with b even and $432 \mid c$, the cited elementary-theta transformation law yields

$$\Theta_{6A_5+v_j}|_{5/2}\gamma = \left(\frac{2\det A}{d}\right) \Theta_{6A_5+pv_j} = \left(\frac{12}{d}\right) \Theta_{6A_5+pv_j}. \quad (3.3)$$

The exponential phase in that formula is one because $(v_j, v_j)/2 \in \mathbb{Z}$. Also $p \equiv \pm 1 \pmod{6}$; the plus sign fixes the coset, while the minus sign gives the same theta series after $z \mapsto -z$. If b is odd, replace γ on the right by $\gamma \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Its upper-right entry is even, and integral Fourier exponents make the theta series translation-invariant. Hence (3.3) extends to every $\gamma \in \Gamma_0(432)$. Positive-definite coset theta theory supplies holomorphy at the cusps, and therefore

$$T \in M_{5/2}(\Gamma_0(432), \chi_{12}). \quad (3.4)$$

Squaring the precise half-integral slash law gives

$$T^2 \in M_5(\Gamma_0(432), \chi_{-4}); \quad (3.5)$$

the residual factor is $\varepsilon_d^{-10} = (-4/d)$.

For G^2 , the eta exponents at 3, 6, 18, 36 are respectively $-4, 6, 4, 4$. The two Newman congruence sums at level 36 are

$$\sum_{\delta|36} \delta r_\delta = 240, \quad \sum_{\delta|36} \frac{36}{\delta} r_\delta = 0,$$

and $\prod_{\delta|36} \delta^{r_\delta} = 6^{18}$. Newman's eta-quotient theorem [4] therefore gives character χ_{-4} . The standard cusp-order formula [3], evaluated at the possible cusp denominators

$$1, 2, 3, 4, 6, 9, 12, 18, 36,$$

gives, in the same order,

$$0, \frac{3}{2}, 0, 2, \frac{3}{2}, 4, 2, \frac{11}{2}, 10.$$

Every order is nonnegative, so $G^2 \in M_5(\Gamma_0(36), \chi_{-4})$, and hence also in the space in (3.5).

It remains to make a finite exact comparison. The index is

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(432)] = 432 \left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{3}\right) = 864,$$

so the weight-5 Sturm bound is $5 \cdot 864/12 = 360$. The coefficients through this bound were computed in two direct integer calculations. For the theta side, $(z, z)/2 \leq 360$ implies $|z_i| \leq 26$; enumerating the six congruence classes $z \equiv v_j \pmod{6A_5}$ in this finite box is therefore exhaustive. For the eta side, (3.1) was expanded as finite truncated products, using $(1-x)^{-2} = \sum_{k \geq 0} (k+1)x^k$. The two coefficient lists agree through q^{360} . As a separate convention check, (2.1) was enumerated through charged degree 118, the last degree that can enter $q^5 C(q^3)$ below the bound; a coordinate of absolute value at least 8 forces the first core size above 118, since $192 - \sqrt{1120} > 118$. This second list also agrees termwise with the coset theta list. The accompanying verifier performs all three operations with exact integers and reports no mismatch.

Sturm's theorem now gives $T^2 = G^2$. Both T and G begin with q^5 , so $T + G \neq 0$. Since formal Laurent series over $\mathbb{C}$ form an integral domain, $(T - G)(T + G) = 0$ implies $T = G$. $\square$

Substituting q^3 into the first product in (1.3) and multiplying by q^5 gives (3.1), so Proposition 3.1 proves the first equality in (1.3). Gauss's products

$$\psi(q) = \frac{(q^2; q^2)_\infty^2}{(q; q)_\infty}, \quad c_3(q) = \frac{(q^3; q^3)_\infty^3}{(q; q)_\infty}$$

give the second equality.

4 Positivity in every degree

We use two elementary representation lemmas.

Lemma 4.1. *For each $z \geq 0$, there is a 3-core partition of size z^2 .*

Proof. In sum-zero 3-abacus coordinates, the size is

$$\frac{3}{2} \sum_{i=1}^3 b_i^2 + \sum_{i=1}^3 i b_i.$$

For $k \geq 0$, use

$$(b_1, b_2, b_3) = \begin{cases} (-k, 2k, -k), & z = 3k, \\ (k+1, k, -2k-1), & z = 3k+1, \\ (-2k-1, k+1, k), & z = 3k+2. \end{cases}$$

Substitution gives z^2 in all three cases. $\square$

Lemma 4.2. *Every $n \geq 0$ can be written*

$$n = x^2 + y(y+1) + 2z^2 \quad (4.1)$$

with $x, y, z \in \mathbb{Z}_{\geq 0}$.

Proof. Put $m = 4n + 1$. By Legendre's three-square theorem, $2m = A^2 + B^2 + C^2$: the integer $2m$ is not of the excluded form $4^a(8b+7)$. Reduction modulo 4 shows that exactly two of A, B, C are odd. Relabel so that A, B are odd and C is even, and put

$$U = \frac{A+B}{2}, \quad V = \frac{A-B}{2}, \quad W = \frac{C}{2}.$$

Then $m = U^2 + V^2 + 2W^2$. Exactly one of U, V is odd, and the congruence $m \equiv 1 \pmod{4}$ forces W to be even. Write the even member of $\{U, V\}$ as $2x$, the odd member as $2y+1$, and $W = 2z$. Changing signs if necessary makes $x, z \geq 0$ and replaces a negative y by $-y-1$. After subtracting 1 and dividing by 4, we obtain (4.1). $\square$

Completion of the proof of Theorem 1.1. Jacobi's identity gives

$$\psi(q)^2 = \phi(q)\psi(q^2).$$

Fix $n \geq 0$ and choose x, y, z as in Lemma 4.2. In

$$c_{6,(0,1)}(q) = \phi(q)\psi(q^2)c_3(q^2)\psi(q^6),$$

select the term q^{x^2} from $\phi(q)$, the term $q^{y(y+1)}$ from $\psi(q^2)$, a 3-core of size z^2 from $c_3(q^2)$ using Lemma 4.1, and the constant term from $\psi(q^6)$. Their product contributes positively to q^n . Every factor has nonnegative coefficients, so cancellation is impossible. This includes $n = 0$. $\square$

5 Scope and reproducibility

Combining Theorem 1.1 with (1.2) gives the exact expression

$$c_{6,(0,3)}(q) = \phi(q)c_3(q)^2 - 2q\psi(q)^2c_3(q^2)\psi(q^6).$$

This is a subtraction, not a nonnegative decomposition; it does not prove positivity for charge $(0, 3)$. Accordingly, this note does not resolve the original conjunction posed by the two $e = 6$ branches.

The finite step in Proposition 3.1 is reproducible with the accompanying scripts `independent_theta_audit.py` and `certify_sturm.py`. They use exhaustive bounded enumeration and integer polynomial arithmetic only; no floating-point test is part of the proof. The product identity was initially suggested by exact coefficient experiments. An AI reasoning and coding system assisted those experiments, the implementation of the exact verifiers, algebraic cross-checks, and manuscript preparation. The theorem rests on the stated deductions and reproducible exact certificate, not on numerical confidence or an originality claim.

References

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